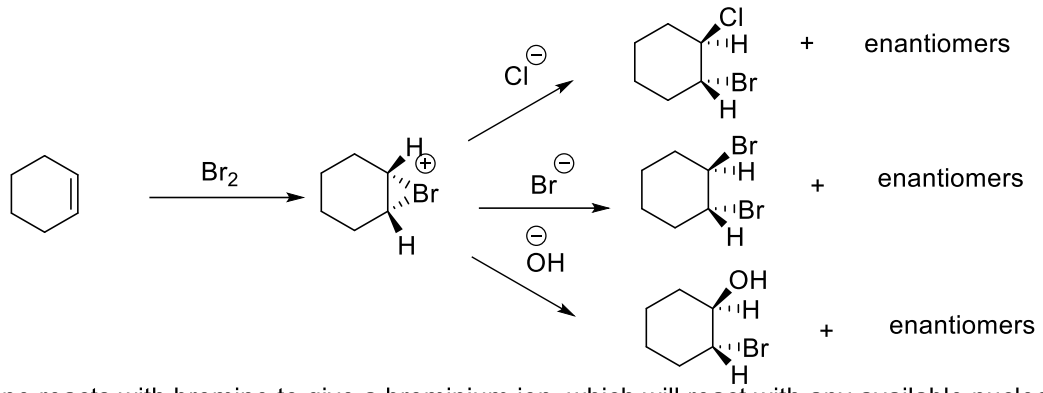
1. When cyclohexene is allowed to react with bromine in an aqueous solution of sodium chloride, the products are trans-1,2-dibromocyclohexane, trans-2-bromocyclohexanol and trans- 1-bromo-2-chlorocyclohexane. Write a plausible mechanism that explain the formation of these products.

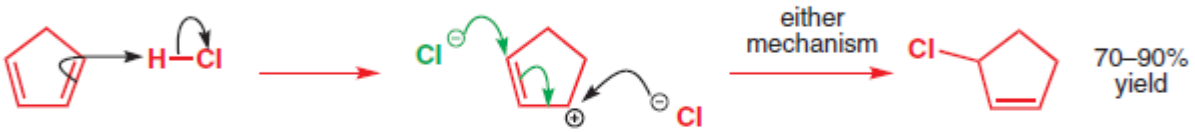


Cyclohexene reacts with bromine to give a brominium ion, which will react with any available nucleophile. The most abundant nucleophile present are Chloride ions, Brominium ions and water.Attack of the respective gives the mixture of results as questioned

2. Predict the product with stereochemistry.

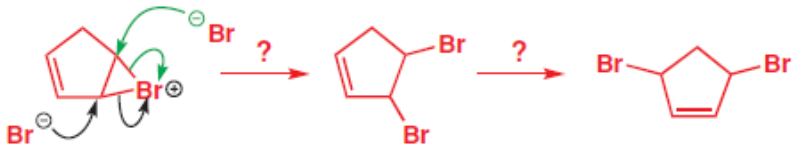
Electrophilic attack on conjugated dienes

Another way to make allylic chlorides is by treating dienes with HCl. Electrophiles attack conjugated dienes more readily than they do isolated alkenes. There was some discussion of this in Chapter 19, establishing the main point that the terminal carbon atoms are the most nucleophilic and that the initial attack produces an allylic cation. A simple example is the addition of HCl to cyclopentadiene.

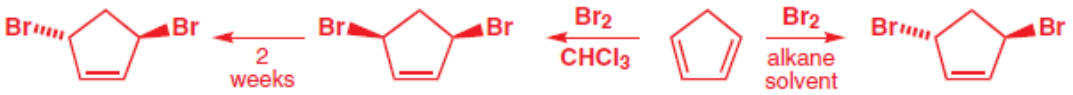


Although there is a question of regioselectivity in the initial protonation, the allylic cation is symmetrical and attack by chloride at either end produces the same product. However, if the electrophile is a halogen rather than HCl or HBr then the reaction becomes regioselective as the cationic intermediate is no longer symmetrical. What happens is this:

The alternative is direct attack on the bromonium ion intermediate, which we assume would occur at the allylic site (black arrows) and not at the other (green arrows). Although this 1,2-dibromide product is not observed, it is still possible that this reaction happens because the 1,2-product can rearrange by bromide shift to the observed 1,4-dibromide.



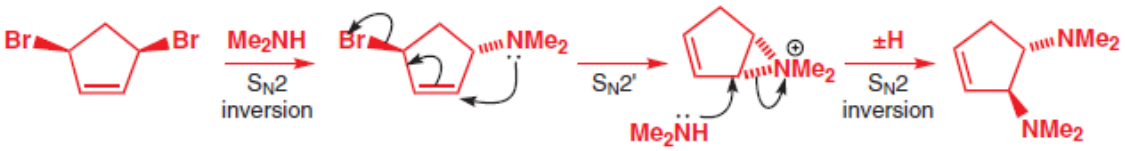
The final product of this reaction could in fact be either of two compounds as the two bromine atoms may be *cis* or *trans*. Bromination in chloroform at –20 °C gives mostly a liquid *cis* dibromide while reaction in hydrocarbon solvents gives the crystalline *trans* isomer. On standing the *cis* isomer slowly turns into the *trans*.



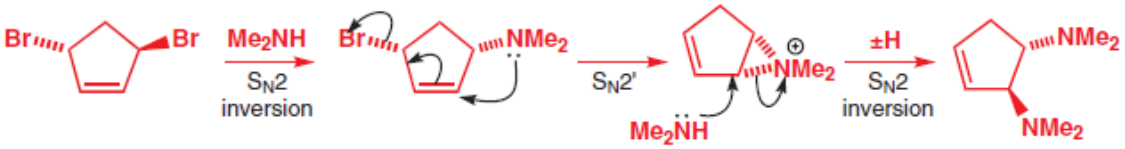
This suggests that the *cis* bromide is the kinetic product and the more stable *trans* compound is the thermodynamic product, formed by reversible loss of bromide and reformation of the bromonium ion.

Similar questions arise when nucleophilic substitution occurs on the dibromides. Reaction of either the *cis* or the *trans* dibromide with dimethylamine gives the *trans* isomer of a diamine. But look at the regioselectivity—it's not the diamine you might expect. The only explanation is one S_N2 displacement and one S_N2' displacement.

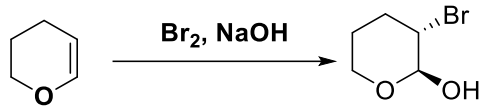
But what about the stereochemistry? Starting with the *cis* isomer, one S_N2 displacement with inversion might be followed by an intramolecular S_N2' displacement and finally another S_N2 displacement with inversion at the allylic centre.



The reaction with the *trans* isomer is almost identical: the same three-membered ring is an intermediate in both sequences so the products are bound to be the same.

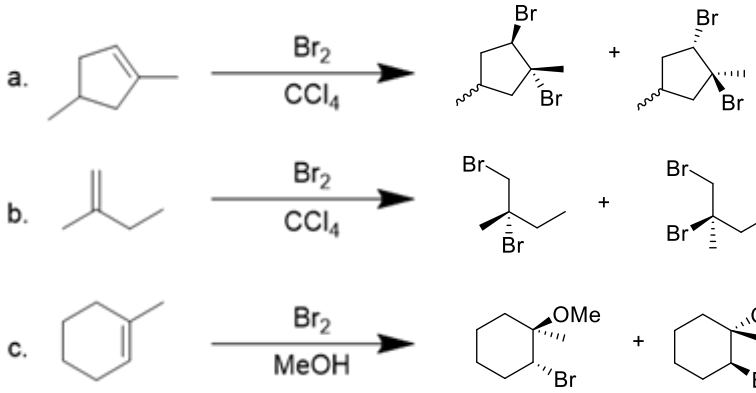


3. The major product formed in the following reaction is:



+ enantiomer

4. Write all possible products (major and minor) of the following reactions with mechanism, also state the relation between these products.

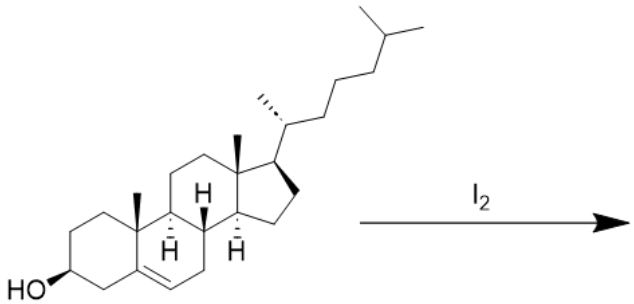


diastereomers

enantiomers

enantiomers

5. State whether the reaction will happen or not, if yes, write the product, if not, mention reason.



$\Delta H = \Delta H$ **(bond broken) +** ΔH **(bond formed)**

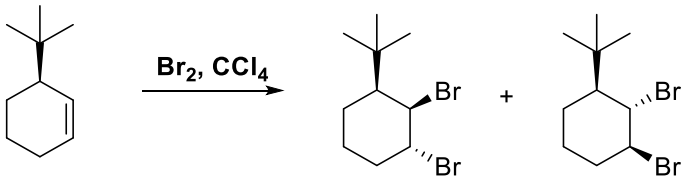
$= \Delta H\ (\pi_{C=C} + \sigma_{I-I}) + \Delta H\ (2 * \sigma_{C-I})$

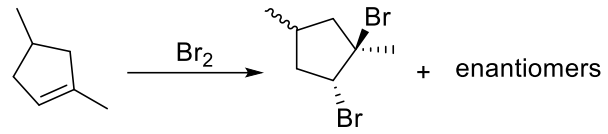
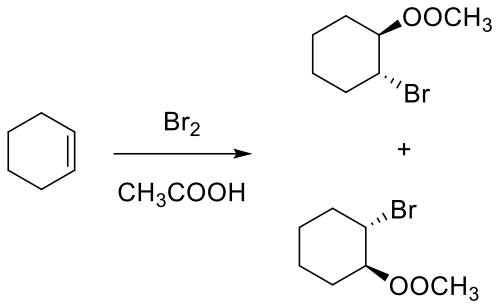
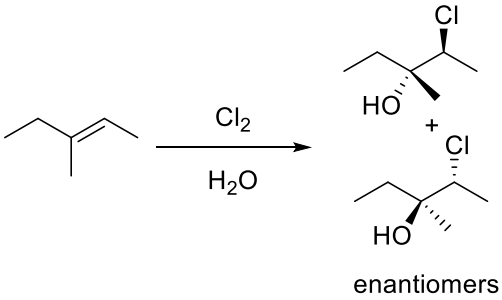
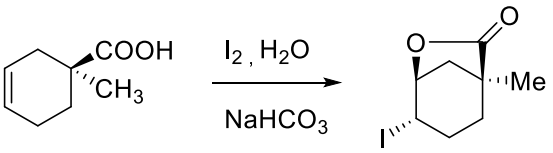
$= +\ (64+3) - (2*51)$

$= -2$ **Kcal/ mol (not sufficient to overcome the entropic penalty)**

$\Delta G = \Delta H - T\Delta S > 0$

6. Predict the product with stereochemistry.



(identification of major isomer will be discussed in 6th Semester)

Class 2_CH2204_Sample Solutions